MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2DSCCSC100 - Python Programming

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

(Answer all questions, each question carries 2 marks.)

1. Name any three key features of Python. [U] [CO1]
2. Identify the output of the following Python expression:

   print(3 * "Python" + "3") [U] [CO1]
3. Which Python data type would you use to store a collection of unique items? [U] [CO3]
4. What is the difference between break and continue statements in Python? [U] [CO2]
5. Rewrite the following for loop as a while loop and ensure it produces the same output:
   fori in range(1, 6):
   print(i * "*") [A] [CO2]

[5 x 2 = 10]

Part B

Short Answer Questions

*(Answer **5** out of 7 questions, each question carries **5** marks.)*

6. Define type conversion in Python. List any four built-in functions used for type conversion. [K] [CO1]
7. Write a Python program that accepts a number as input and prints whether it is positive, negative, or zero using if-elif-else. [A] [CO2]
8. Explain the concept of slicing in Python lists with an example. [U] [CO3]
9. Describe the working of a for loop with an example that prints the first 10 even numbers. [U] [CO2]
10. Write a Python function to calculate and return the factorial of a given number. [A] [CO3]
11. Discuss the difference between Lists and Tuples with examples. [U] [CO3]

12. Write a Python script to open a file, write a few lines of text into it, and then read its content. [A] [CO3]

[5 x 5 = 25]

Part C

Essay Questions

*(Answer **1** out of 2 questions, each question carries **15** marks.)*

13. Explain the concept of dictionaries in Python. Discuss various dictionary methods with examples. How is the del statement used to delete dictionary elements? Provide relevant code snippets to support your explanation. [U] [CO3]

14. Explain the concept of sets in Python. How do you create a set? Discuss various set methods with examples. Provide relevant code snippets to demonstrate set operations. [U] [CO3]

[1 x 15 = 15]